

1.15 (a) 对 S_1 : $y_1[n] = 2x[n] + 4x[n-1]$

对 S_2 : $y_2[n] = y_1[n-2] + \frac{1}{2}y_1[n-3] = 2x[n-2] + 4x[n-3] + \frac{1}{2}(2x[n-3] + 4x[n-4])$
 $= 2x[n-2] + 5x[n-3] + 2x[n-4]$

(b) 若交换, 对 S_2 : $y_2[n] = \cancel{2x[n-2] + 4x[n-3]} + \frac{1}{2}x[n-3]$

对 S_1 : $y_1[n] = 2x[n-2] + 4x[n-3] + \frac{1}{2}(2x[n-3] + 4x[n-4])$
 $= 2x[n-2] + 5x[n-3] + 2x[n-4]$ 不会发生变化.

1.16 $y[n] = x[n]x[n-2]$

(a) 不是 (b) $x[n], x[n-2]$ 中必有 ≥ 1 为 0, 所以 $y[n] = 0$.

(c) 不是

	① memoryless	② time invariant	③ linear	④ Causal	⑤ Stable
1.27 (a) $y(t) = x(t-2) + x(2-t)$	X	X	✓	X	✓
(b) $y(t) = [\cos 3t] \cdot x(t)$	✓	X	✓	✓	✓
(c) $y(t) = \begin{cases} 0, & x(t) < 0 \\ x(t) + x(t-2), & x(t) \geq 0 \end{cases}$					
(c) $y(t) = \int_{-\infty}^{2t} x(\tau) d\tau$	X	X	✓	X	X
(d) $y(t) = \begin{cases} 0, & \cancel{x(t)} \leq t < 0 \\ x(t) + x(t-2), & t \geq 0 \end{cases}$	X	X	✓	✓	✓
(e) $y(t) = \begin{cases} 0, & x(t) < 0 \\ x(t) + x(t-2), & x(t) \geq 0 \end{cases}$	X	✓	X	✓	✓
(f) $y(t) = x(\frac{t}{3})$	✓ X	X	✓	X $n < 0$ 时	✓
(g) $y(t) = \frac{dX(t)}{dt}$	X	✓	✓	✓	X $\lim_{t \rightarrow \infty} X(t) = \cos t$
1.28 (a) $y[n] = x[-n]$	X	X	✓	X $n < 0$ 时	✓
(c) $y[n] = nx[n]$	✓	X	✓	✓	X $\lim_{n \rightarrow \infty} y[n] = \infty$
(d) $y[n] = \sum \{x[n-1]\}$ $\Rightarrow y[n] = \frac{1}{2} \{x[n-1] + x[-n-1]\}$	X $n \geq 0$ 时, $\cancel{x[n-1]}$	X	✓	X $n < -n-1$ 时	✓
(g) $y[n] = x[4n+1]$	X	X	✓	X	✓

1.31 (a) $x_1(t)$:

$y_1(t)$:

$x_2(t)$:

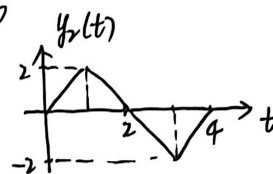
$$x_2(t) = x_1(t) - x_1(t-2)$$

$$\Rightarrow y_2(t) = y_1(t) - y_1(t-2)$$

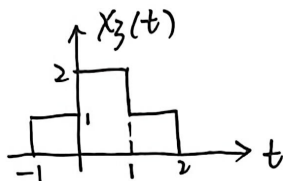
$t \in (-\infty, 0)$ 时, $t-2 < 0$, $y_1(t) = y_1(t-2) = 0$, $y_2(t) = 0$

$t \in (0, 2)$ 时, $t-2 < 0$, $y_1(t-2) = 0$, $y_2(t) = y_1(t)$

$t \in (2, 4)$ 时, $y_1(t) = 0$, $y_2(t) = -y_1(t-2)$

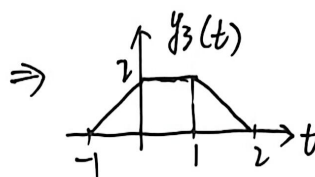


(b)

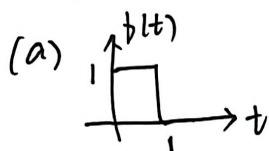


$$x_3(t) = x_1(t) + x_1(t+1)$$

$$\Rightarrow y_3(t) = y_1(t) + y_1(t+1)$$



Problem 6.



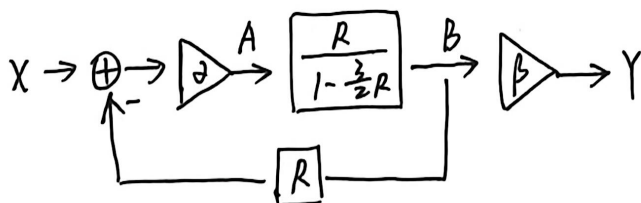
$$x_c(t) = \sum_{n=-\infty}^{\infty} x_d[n] \cdot b\left(\frac{t}{T} - n\right)$$

(b) $y_c(t) = \sum_{n=-\infty}^{\infty} [y_d[n] \cdot c\left(\frac{t}{T} - n\right) + y_d[n+1] \cdot a\left(\frac{t}{T} - n\right)]$



(c) $\frac{dy_c(t)}{dt} = \sum_{n=-\infty}^{\infty} \frac{1}{T} (y_d[n+1] - y_d[n]) \cdot b\left(\frac{t}{T} - n\right)$

Problem 7.



$$\begin{cases} A = 2(X - RB) \\ B = A \cdot \frac{R}{1 - \frac{3}{2}R} \end{cases} \Rightarrow B = 2(X - RB) \cdot \frac{R}{1 - \frac{3}{2}R}, B = \frac{2R \cdot X}{1 - \frac{3}{2}R + 2R^2}$$

$$Y = \beta B = \frac{2\beta R \cdot X}{1 - \frac{3}{2}R + 2R^2}$$

$$\text{令 } Y(R) = \sum y[n] R^n,$$

$$\begin{cases} Y(R) = 0 + R + \frac{3}{2}R^2 + \frac{7}{4}R^3 + \dots \\ RY(R) = R^2 + \frac{3}{2}R^3 + \frac{7}{4}R^4 + \dots \end{cases}$$

$$\Rightarrow (1-R)Y(R) = R + \frac{1}{2}R^2 + \frac{1}{4}R^3 + \dots = R \cdot \sum_{n=0}^{\infty} \frac{1}{2^n} \cdot R^n = R \cdot \frac{1}{1-\frac{R}{2}}$$

$$\Rightarrow Y(R) = \frac{R}{(1-R)(1-\frac{R}{2})}$$

$$\because X[n] = \delta[n] \quad \therefore \frac{\partial \beta R}{1 - \frac{3}{2}R + 2R^2} = \frac{R}{(1-R)(1-\frac{R}{2})}$$

$$\Rightarrow \begin{cases} \alpha = \frac{1}{2} \\ \beta = 2 \end{cases}$$